Viva preparation: questions, calculations and contribution

Micro heat pipes and vapour chambers · 3-minute talk with a 2-minute Q&A · Debanshu Ghosh, 23CH3FP56. Written for you, the presenter: read Part A the night before, use Part B when someone asks “how did you get that number?”, and know Part D by heart.

Part	What it gives you
A	25 likely questions with answers. The first nine (★, shaded) are the ones most likely to be asked. Each answer is sized to fit in 20 seconds.
B	How every number on the slides is calculated: inputs, formulas, worked arithmetic, tables.
C	What came from research papers, what is textbook theory, and what was assumed.
D	Your contribution, what is not new, and one sentence to say it in.

How to survive a 2-minute Q&A

- Expect 3 to 5 questions. Answer in 15 to 20 seconds: the number or the one-line reason first, the explanation second.
- Your numbers are for water at 60 °C and an illustrative 5 W phone-like disc. Say “in my model” before a number. Nobody can fault that; they can fault a number presented as a measurement.
- If you do not know: say what you do know, then “that is outside my model; I would check it in the literature.” Do not guess a value.
- Easy exits: “that is an assumption, stated on the slide”; “that is an upper bound, not a prediction”; “that is a simulation result from Patankar, not mine”.
- Slides 3, 4 and 5 hold your three biggest numbers: 5× and 25×; 50, 38 and 161 W; 27×, 28×, 12× and about 2×.

Part A · Questions and answers

The big picture

Q1. ★ How does a vapour chamber move heat without a pump?

Heat evaporates water in the wick at the chip. The vapour flows through the open core to a cooler area and condenses, releasing the latent heat. The condensed liquid returns through the porous wick by capillary suction. The wick is the only pump, so the capillary pressure must beat the liquid loss, the vapour loss and gravity. That sets a maximum heat load, the capillary limit.

If pushed: In equations: $\Delta P_{cap} \geq \Delta P_l + \Delta P_v + \Delta P_g$, and since both flow losses are linear in Q, $Q_{max} = (\Delta P_{cap} - \rho_l g \Delta z)/(a_l + a_v)$. My two topics are the liquid term and the vapour term.

Q2. What is the capillary limit, and what happens if you exceed it?

It is the largest heat load for which the capillary pressure can still return enough liquid. Beyond it the wick cannot resupply the evaporator, the wick dries out near the heat source, and the temperature there rises sharply.

If pushed: There are other limits (sonic, entrainment, boiling). They are outside my scope. As a check, the sonic limit in my geometry is far above the 5 W load: about 228 W at h = 100 µm.

Q3. What is the difference between a micro heat pipe, a heat pipe and a vapour chamber?

They are the same physics in different shapes. A heat pipe moves heat along a line, from an evaporator end to a condenser end. A vapour chamber is the flat, two-dimensional version that spreads heat over a plane. A micro heat pipe is a very small one, where the corners of the channel carry the liquid. My numbers use the flat chamber geometry.

Q4. Why water, and why 60 °C?

Water has a high latent heat (2.358 MJ/kg), a high surface tension (0.0662 N/m) and is compatible with copper, so it gives a strong pump and carries a lot of heat per kilogram. 60 °C is an illustrative working temperature for electronics cooling; the properties change only modestly with temperature, so the trends do not depend on it.

If pushed: I did not run a temperature sweep, so do not claim a precise dependence. The numbers would shift; the 1/d, d^2 and $1/h^3$ scalings would not.

Topic 1: the wick (liquid return, biporous)

Q5. ★ Why is a finer wick not always better?

Because one length sets both the pump and the resistance. Capillary pressure goes as 1/r, so it rises as pores shrink. But permeability goes as d^2, so the liquid resistance rises faster. Going from 50 µm to 10 µm powder gives 5× the pump pressure but 25× the liquid resistance. For the liquid alone, $Q \propto \Delta P_{cap} \cdot K \propto d$, so finer actually delivers less.

If pushed: In the full model with a 100 µm core, the 50 µm powder reaches about 56 W and the 10 µm powder only about 29 W, held back by its own wick.

Q6. ★ What is a biporous wick, and where does 161 W come from?

It has two pore scales: fine pores inside particle clusters set the pump pressure, and coarse channels between the clusters carry the liquid. In my model I take the pressure from r = 5 µm and the permeability from a 100 µm powder. That gives 161 W against 50 W for the fine wick and 38 W for the coarse one. It is an upper bound: it assumes the two regions do not interfere.

If pushed: Why upper bound: the channels take up volume and break the conduction path, and K from 100 µm particles would not fit in a 50 µm layer. The bar shows the principle, not a buildable wick.

Q7. ★ What is the 37% on your biporous slide? Is it your result?

No, it is not mine. It is from Patankar, Weibel and Garimella (2016): a 3-D numerical simulation in which a biporous condenser-side wick allowed a thicker vapour core, and the peak-to-mean surface temperature difference on the condenser fell by 37% compared with a monolithic wick.

If pushed: Be exact about it: simulation, not experiment; condenser surface temperature difference, not chip temperature; relative to a monolithic wick. Semenic and Catton (2009) is the experimental work on biporous wicks for evaporators.

Q8. How did you get the permeability K?

From the Kozeny–Carman relation for a bed of spheres: $K = d^2\varepsilon^3/[150(1-\varepsilon)^2]$. With porosity $\varepsilon = 0.5$ it is $K = d^2/300$. For d = 50 µm that is 8.33e-12 m². The mesh value, $K = 2.8\times10^{-11}$ m², is a typical figure I chose, not computed.

If pushed: Kozeny–Carman treats the pores as tubes with the hydraulic diameter of a packed bed. It is an estimate, and real wicks differ from it.

Q9. Where does $r_{eff} = 0.21\ d$ come from?

It is a common rule of thumb for sintered spheres: the effective pore radius is about a fifth of the particle diameter. I state it as an assumption; it is not derived in my work.

If pushed: Mesh and groove wicks have their own correlations (Peterson 1994, Faghri 1995).

Q10. Is Darcy's law valid for a 50 µm wick?

Only roughly. Darcy's law and Kozeny–Carman are continuum descriptions that need a layer many particles thick. A 50 µm layer of 50 µm powder is about one particle deep, so that case is an order-of-magnitude estimate.

If pushed: It does not change the trend, because the 5× and 25× come from scalings (1/d and d²). It does affect the absolute watts.

Q11. What if the wick does not wet perfectly?

The capillary pressure carries cos θ. A contact angle drifting from 0° to 60° through oxidation or contamination halves the pump pressure, with no change in the geometry.

Q12. Does orientation matter, for example a phone held upright?

In the model, only through the gravity head $\rho_l\, g\, \Delta z$. For a 100 mm rise that is 0.96 kPa. It would cut my idealised biporous limit from 161 W to 155 W if the source is at the top of a vertical sheet, while the fine wick, with its high pump pressure, is hardly affected.

If pushed: At the sub-millimetre scale the Bond number is 0.006 at 200 µm, so inside the sheet surface tension wins over gravity.

Topic 2: the vapour core

Q13. ★ Why does the vapour pressure drop go as $1/h^3$?

Plane Poiseuille flow in a slit gives a pressure gradient $12\mu u/h^2$. The flow area is $2\pi rh$, so the velocity itself goes as $1/h$. One power of h comes from the velocity, two from the wall shear: $1/h \times 1/h^2 = 1/h^3$. So halving the gap multiplies the pressure drop by $2 \times 4 = 8$.

If pushed: Integrating from R1 to R2 gives $\Delta P_v = 6\, \mu_v\, \dot{m}\, \ln(R2/R1)/(\pi\, \rho_v\, h^3)$. At 5 W and h = 100 µm that is 718 Pa; at 50 µm it is 5745 Pa, which is 8 times larger.

Q14. ★ How does a pressure drop become a temperature drop?

In saturated vapour, pressure and temperature are tied by Clausius–Clapeyron: $\Delta T \approx T\, \Delta P/(\rho_v\, h_{fg})$. For water at 60 °C that is about 1.1 K for every kilopascal of vapour pressure drop.

If pushed: Example: 718 Pa at 100 µm and 5 W is 0.78 K. The relation is linearised, so it is only valid while the drop is a small fraction of P_{sat}; I use 20% (about 56 µm at 5 W) as the limit.

Q15. ★ What does your copper comparison show, and how is it set up?

A solid copper sheet of the same total thickness carries the same 5 W from the same source to the same rim. At 5 W the chamber has a temperature drop 27 times smaller at h = 400 µm, 28 times at 200 µm, 12 times at 100 µm and only about 2 times at 50 µm. Thinning the core erases the advantage, because copper degrades slowly with thickness while the vapour drop grows as $1/h^3$.

*If pushed: The chamber's drop is vapour + wick conduction + evaporating interface. The interface term assumes an accommodation coefficient of 0.1. The 50 µm value is marked * because the vapour drop there is above 20% of P_{sat}, so read it as a trend.*

Q16. Why is your 50 µm chamber value marked with an asterisk?

At 50 µm the vapour pressure drop at 5 W is 5.7 kPa, which is above 20% of the saturation pressure (19.95 kPa). The linearised Clausius–Clapeyron relation is then no longer valid, so that value is a trend, not a prediction.

Q17. Is the vapour flow incompressible and continuum?

Yes in the range used. At h = 100 µm and 5 W the Mach number is 0.010 and the Knudsen number is 0.0077 (mean free path 0.77 µm), so the flow is incompressible and the continuum, no-slip description holds.

Q18. What are the main assumptions behind the vapour-core result?

Steady, laminar, incompressible flow; smooth walls; and all the vapour travelling to the rim R2. In reality vapour condenses along the way, so the true pressure drop is lower by a factor of order one. Support pillars would add drag and raise it.

If pushed: So my number is conservative on the first point and optimistic on the second. I use the model for the $1/h^3$ trend, not for the exact watts.

Q19. You removed the thickness-optimisation topic. What was it, in case I ask?

With a fixed inner thickness of 250 µm, every micron of wick comes out of the vapour core. Liquid resistance falls as 1/t and vapour resistance rises as $1/h^3$, so the total has a minimum. Setting the derivative to zero gives $h^2/t = 12\sqrt{(K\nu_v/\nu_l)}$. For the 50 µm powder: t ≈ 50 µm, h ≈ 151 µm, about 101 W.

If pushed: It is in the written paper. The result is my own derivation from the model, checked against a numerical sweep, and not compared with a published optimum. It maximises the capillary limit only, not thermal resistance.

Applications

Q20. ★ Why heat pipes in satellites, and what is special about gravity there?

In orbit there is no gravity, so the liquid cannot drain back; the wick alone must return it. A heat pipe does that with no moving parts and moves heat with very little temperature drop. In my pressure budget the gravity term ΔP_g is then zero, so the capillary pressure only has to beat the liquid and vapour losses.

If pushed: On Earth the term is $\rho_l\, g\, \Delta z$, which is 0.96 kPa for a 100 mm rise. At the sub-millimetre scale of a phone chamber surface tension dominates: the Bond number at 200 µm is only 0.006.

Q21. Why heat pipes in laptops but a vapour chamber in a phone?

A laptop has room to run a heat pipe from the CPU to a fan and fins, so the job is moving heat along a line. A phone has no fan and under a millimetre of free thickness, so the job is spreading the heat across the whole body, which a flat chamber does.

Method and honesty

Q22. ★ Are your numbers from a real device? Is the model validated?

No. They come from a deliberately simple model: saturated water at 60 °C, a disc-shaped chamber with a 10 × 10 mm source and a 48 mm rim, 5 W. They are illustrative, not measured, and I have not validated the model against experiments. The model is there to show trends and scalings, and to say which effect dominates.

If pushed: Published experiments are cited only as context. For example, Yan et al. (2023) report a 0.5 mm composite-wick chamber with critical powers up to 60 W; that device differs from my model, so it is not a validation.

Contribution and limits

Q23. What is your own contribution here?

A transparent first-principles model that puts the wick and the vapour core into one capillary-limit equation and quantifies them for a phone-sized chamber: the 5× and 25× wick trade-off, the idealised biporous bound (50, 38 and 161 W), the $1/h^3$ vapour limit with its validity checks, and the comparison with a copper sheet. I did not run experiments, and the formulas themselves are standard.

If pushed: See Part D for the full answer, including what is not new.

Q24. What would you do next?

Validate the trends against a published experiment or a CFD model; add support pillars and several heat sources; replace the Darcy estimate for the thin wick with a pore-scale model; and run the temperature sweep.

Q25. What is the weakest part of your analysis?

That it is unvalidated and that some inputs are assumptions: the 5 W load, the geometry, the 0.21 d rule, the mesh permeability and the accommodation coefficient of 0.1. The trends are robust, because they come from scalings; the absolute watts and kelvins are not.

Part B · How the numbers are calculated

Everything on the slides comes from one script (calc.py, with its companion script for the extra steps), using standard steam-table properties of saturated water at 60 °C and the geometry in B1. The chain is always the same: **inputs → formula → number → check**. Below, each step shows the formula, one worked calculation you can repeat on a calculator, and the table the slide uses.

B1. Inputs

Quantity	Value	Kind of input
Surface tension σ	0.0662 N/m	standard steam-table value
Latent heat h_{fg}	2.358 MJ/kg	standard steam-table value
Liquid density, viscosity	983.2 kg/m³, 4.66×10^{-4} Pa s	standard steam-table values
Vapour density, viscosity	0.1304 kg/m³, 1.08×10^{-5} Pa s	standard steam-table values
Saturation pressure P_{sat}, T	19.95 kPa, 333.15 K (60 °C)	standard steam-table values
Heat load Q, source	5 W on 10 × 10 mm, so R1 = √(A/π) = 5.64 mm	assumed (phone-like, illustrative)
Chamber	disc with R2 = 48 mm, the same area as a 60 × 120 mm sheet	assumed
Wick layers, copper walls	2 × 50 µm wick, 2 × 80 µm wall	assumed
Porosity ε, r_{eff}	ε = 0.5; $r_{eff} \approx 0.21$ d for sintered powder	assumed; rule of thumb
Cu mesh #200	r_{eff} = 64 µm, $K = 2.8\times10^{-11}$ m²	assumed typical values
Contact angle	θ = 0° (full wetting)	assumed
Gravity head	Δz = 100 mm	assumed (vertical case)

Derived once and reused: ln(R2/R1) = ln(48/5.642) = 2.141, and the mass flow at 5 W, $\dot{m} = Q/h_{fg} = 5/2.358\times10^{6} = 2.120\text{e-}06$ kg/s.

B2. The pump: capillary pressure (slide 3)

$$\Delta P_{cap} = \frac{2\sigma\cos\theta}{r_{eff}}$$

Worked: 50 µm powder, $r_{eff} = 0.21 \times 50 = 10.5$ µm, so $\Delta P_{cap} = 2 \times 0.0662 / 10.5\times10^{-6} =$ 12.61 kPa. 10 µm powder: $r_{eff} = 2.1$ µm gives 63.0 kPa. Ratio = 5 (it is 50/10 = 5, because $\Delta P_{cap} \propto 1/d$).

B3. The loss: permeability and liquid resistance (slide 3)

$$K = \frac{d^2\varepsilon^3}{150(1-\varepsilon)^2}$$

$$\Delta P_l = a_l\, Q, \qquad a_l = \frac{\mu_l \ln(R_2/R_1)}{2\pi\, \rho_l\, K\, t_l\, h_{fg}}, \qquad t_l = 2t$$

Worked: 50 µm powder, $K = (50\times10^{-6})^2 \times 0.125 / (150 \times 0.25) = 8.333\text{e-}12$ m² (that is $d^2/300$). With $t_l = 100$ µm: $a_l = 4.66\times10^{-4} \times 2.141 / (2\pi \times 983.2 \times 8.333\text{e-}12 \times 1\times10^{-4} \times$

2.358×10⁶) = 82.2 Pa/W. The same steps for 10 µm powder give 2055 Pa/W, a ratio of 25 (K ∝ d², so 25).

Wick	r_{eff} (µm)	K (m²)	ΔP_{cap} (kPa)	a_l (Pa/W)
Cu mesh (#200)	64.0	2.80e-11	2.1	24.5
Sintered, d = 50 um	10.5	8.33e-12	12.6	82.2
Sintered, d = 10 um	2.1	3.33e-13	63.0	2054.7

B4. The vapour resistance and its temperature cost (slide 5)

$$\Delta P_v = a_v\, Q, \qquad a_v = \frac{6\,\mu_v \ln(R_2/R_1)}{\pi\,\rho_v\, h^3\, h_{fg}}, \qquad \Delta T_{sat} \approx \frac{T \Delta P}{\rho_v\, h_{fg}}$$

Worked: h = 100 µm, a_v = 6 × 1.08×10⁻⁵ × 2.141 / (π × 0.1304 × (10⁻⁴)³ × 2.358×10⁶) = 144 Pa/W. At 5 W: ΔP_v = 718 Pa. The conversion factor is T/(ρ_v h_{fg}) = 333.15 / (0.1304 × 2.358×10⁶) = 1.08×10⁻³ K/Pa, so ΔT = 0.78 K. Halving h to 50 µm multiplies a_v by 2³ = 8.

h (µm)	ΔP_v at 5 W (Pa)	ΔT at 5 W (K)	Mach	Knudsen	Valid? (ΔP < 20% P_{sat})
400	11	0.01	0.003	0.0019	yes
200	90	0.10	0.005	0.0038	yes
100	718	0.78	0.010	0.0077	yes
50	5745	6.22	0.020	0.0154	no: trend only

The 20% limit is ΔP_v = 0.2 × 19.95 kPa = 3.99 kPa; solving a_v Q = 3.99 kPa for h at 5 W gives h ≈ 56 µm. Below that the linearised relation is not trusted. Validity checks at h = 100 µm: Mach = u/√(γ R_v T) = 0.010 (u = 4.6 m/s, incompressible); Knudsen = λ/h = 0.77/100 = 0.0077 (continuum, no slip).

B5. The capillary limit and the biporous bars (slide 4)

$$Q_{max} = \frac{\Delta P_{cap} - \rho_l g\, \Delta z}{a_l + a_v}$$

Worked, 50 µm powder with h = 100 µm (horizontal, Δz = 0): Q_{max} = 12610 / (82.2 + 144) = 56 W. The three biporous bars use the same equation with a different (r_{eff}, K) pair, all at h = 100 µm and 2 × 50 µm wick:

Case	r for ΔP_{cap}	K for a_l	ΔP_{cap}	a_l	a_v	Q_{max}
Fine only	5 µm	1.76e-12 (d = 23 µm)	26.5 kPa	388 Pa/W	144	50 W
Coarse only	21 µm	3.33e-11 (d = 100 µm)	6.3 kPa	20.5 Pa/W	144	38 W
Biporous	5 µm	3.33e-11	26.5 kPa	20.5 Pa/W	144	161 W

Check the biporous bar: 26480 / (20.5 + 144) = 161 W. The fine wick takes its permeability from a 23 µm powder because its pore radius is 0.21 × 23 ≈ 5 µm, so r and K are consistent with each other. The biporous bar combines the fine r with the coarse K, which is why it is an idealised upper bound.

B6. Chamber against a solid copper sheet (slide 5)

$$\Delta T_{Cu} = \frac{Q \ln(R_2/R_1)}{2\pi\, k\, H}, \qquad \Delta T_{chamber} = \Delta T_{vapour} + \Delta T_{wick} + \Delta T_{interface}$$

The copper sheet has the chamber's total thickness H = h + 2 × 50 + 2 × 80 µm and k = 400 W/m K. The chamber terms: vapour from B4; wick conduction $\Delta T_w = q'' t_w / k_{wick}$ with q″ = Q/A = 50000 W/m² and k_{wick} = 160 W/m K from the Maxwell estimate for ε = 0.5; interface ΔT_i =

q''/h_i with $h_i = 2.33\times10^5$ W/m² K from kinetic theory (Hertz–Knudsen–Schrage) with an assumed accommodation coefficient of 0.1.

Worked, h = 200 µm: H = 0.46 mm, $\Delta T_{Cu} = 5 \times 2.141 / (2\pi \times 400 \times 0.00046) = 9.3$ K. Chamber: 0.097 + 0.016 + 0.215 = 0.33 K. Ratio = 28.

h (µm)	H (mm)	Copper (K)	Vapour (K)	Wick (K)	Interface (K)	Chamber (K)	Copper ÷ chamber
400	0.66	6.5	0.012	0.016	0.215	0.24	27×
200	0.46	9.3	0.097	0.016	0.215	0.33	28×
100	0.36	11.8	0.778	0.016	0.215	1.01	12×
50	0.31	13.7	6.224	0.016	0.215	6.45*	≈2×*

** ΔP_v above 20% of P_{sat}, so a trend only. Why copper catches up: copper's ΔT rises only from 6.5 to 13.7 K as H falls from 0.66 to 0.31 mm, while the chamber's vapour term rises from 0.01 to 6.2 K because it goes as 1/h³.*

B7. The thickness optimum (written paper only)

Not in the 3-minute deck, but it is in the paper, so you may be asked. With H = 2t + h fixed, the total flow resistance is $a_l + a_v = C_l/t + C_v/h^3$. Differentiating with dh/dt = −2 and setting it to zero gives $h^4/t^2 = 6C_v/C_l$, which simplifies to $h^2/t = 12\sqrt{(K\nu_v/\nu_l)}$, where $\nu = \mu/\rho$.

Wick	h^2/t (µm)	t (µm)	h (µm)	Q_{max} (W)	sweep check (t)
Cu mesh #200	839	36.9	176	35	36 µm
Sintered, d = 50 µm	458	49.6	151	101	50 µm
Sintered, d = 10 µm	92	81.7	87	43	82 µm

The closed form lands on the numerical sweep within the 2 µm grid for all three wicks. It is derived here from the model; it has not been compared with a published optimum.

Part C · What came from where

Three kinds of input go into the work, and an examiner is entitled to ask which is which. The honest split is below.

C1. Numbers taken from research papers

Number or claim	Where it appears	Source	Note
37% lower peak-to-mean condenser surface ΔT with a biporous condenser wick	slide 4 tile	Patankar, Weibel & Garimella (2016), Int. J. Heat Mass Transfer 101, 927–936	a 3-D simulation, not an experiment; relative to a monolithic wick
Up to 34% performance reduction by thermal throttling on a Nexus 6P	paper introduction (not on the slides)	Bhat, Gumussoy & Ogras (2019), DATE 2019	motivation only
0.5 mm composite-wick chamber, critical power up to 60 W	paper, context	Yan et al. (2023), Applied Thermal Engineering 226, 120340	context, not a validation of the model
Biporous wicks help high-flux evaporators (experiments)	slide 4 and paper; qualitative	Semenic & Catton (2009), IJHMT 52, 5113–5121	no number from this paper is used
Effective conductivity falls as support-column diameter grows	paper, one sentence	Zhang et al. (2024), Micromachines 15, 627	not on the slides
Vapour chambers move heat as latent heat, with no moving parts	paper introduction	Faghri (2012); Weibel & Garimella (2013); Zhou et al. (2023)	review papers

Only the 37% is a quantitative result borrowed from a paper that appears on the slides. Every other number on the slides comes from the model in Part B.

C2. Theory taken from textbooks and the course (not new, not mine)

Result	Used for
Young–Laplace: pressure jump 2σ/R across a curved interface	capillary pressure
Darcy's law, radial form	liquid pressure loss in the wick
Kozeny–Carman permeability for a bed of spheres	K of the sintered wicks
Plane Poiseuille flow between plates	vapour pressure loss in the core
Clausius–Clapeyron relation	converting vapour pressure drop to temperature drop
Maxwell effective conductivity of a porous medium	wick conduction term in the copper comparison
Hertz–Knudsen–Schrage kinetic theory of evaporation	interface term in the copper comparison
Mach and Knudsen number criteria; Bond number	validity checks
Course material (DasGupta & Ganguly, Lectures 1–4)	scaling, continuum and slip arguments as background
Peterson (1994), Faghri (1995)	standard textbooks cited for classical wick correlations

C3. Values that are assumptions, not data

- **Properties of water at 60 °C**: standard saturated-water steam-table values, rounded. The paper does not name one specific table, so if asked say “standard steam tables”.
- **The operating point**: 5 W, a 10 × 10 mm source, a 48 mm rim, 2 × 50 µm wicks, 2 × 80 µm walls, 100 mm gravity head. All are illustrative and were chosen to be phone-like.
- **$r_{eff} \approx 0.21\ d$ and $\varepsilon = 0.5$** for sintered powder: a rule of thumb and a typical porosity. Not derived here, and not cited to one specific paper.
- **Cu mesh #200 ($r = 64$ µm, $K = 2.8\times10^{-11}$ m²)**: typical values chosen for the example, not computed here.
- **Accommodation coefficient 0.1** in the interface term: assumed. The literature gives a very wide range of values for water (the full paper lists Marek & Straub, 2001, on this).
- **The biporous pair** (r_{fine} = 5 µm, K from a 100 µm powder): chosen to make the principle visible, which is why the result is an upper bound.
- **The 20% of P_{sat} limit** for the linearised Clausius–Clapeyron relation: a judgement, chosen as a conservative cut-off.
- **Copper conductivity 400 W/m K**: a typical handbook value.

Part D · What your contribution is

D1. What you did

- **Framed the three topics as terms of one equation.** The pressure budget $\Delta P_{cap} \geq \Delta P_l + \Delta P_v + \Delta P_g$ and $Q_{max} = (\Delta P_{cap} - \rho_l g \Delta z)/(a_l + a_v)$ tie liquid return, thickness and vapour flow together, so each result has a place.
- **Integrated the radial flow laws for a disc.** The Darcy and Poiseuille laws were integrated from the source radius R1 to the rim R2, giving the ln(R2/R1) forms and, through them, flow resistances a_l and a_v in Pa/W.
- **Quantified the wick trade-off**: 5× the pump pressure but 25× the liquid resistance from 50 µm to 10 µm powder, and the result that for liquid alone $Q \propto d$.
- **Built the idealised biporous bound**: 50, 38 and 161 W at h = 100 µm, with the limits of the idea stated.
- **Worked out the 1/h³ vapour limit with its checks**: ΔT of about 1.1 K per kPa, Mach and Knudsen numbers, and the h below which the linear model fails (about 56 µm at 5 W).
- **Compared the chamber with an equal-thickness copper sheet**: 27×, 28×, 12× and about 2×, showing that a thin core erases the advantage.
- **Derived the thickness optimum** $h^2/t = 12\sqrt{(K\nu_v/\nu_l)}$ (paper only), and checked it against a numerical sweep: t ≈ 50 µm, h ≈ 151 µm, about 101 W for the 50 µm powder.
- **Kept the whole chain reproducible**: one script produces every number, table and figure in the paper and the slides.

D2. What is not new, so do not claim it

- The governing equations (Young–Laplace, Darcy, Poiseuille, Kozeny–Carman, Clausius–Clapeyron) are standard.
- The idea of a biporous wick and the 37% condenser result are from the literature (Semenic & Catton; Patankar et al.).
- There are no experiments. The model is not validated, and its inputs are illustrative. It shows scalings and trends, not device performance.
- The h^2/t closed form is your derivation inside the model. it has not been compared with a published optimum, so do not say it is “new”; say it is “derived here and checked against a numerical sweep”.

D3. One sentence to say it in

“My contribution is a simple, transparent model that puts the wick and the vapour core into one capillary-limit equation, and uses it to show for a phone-sized chamber that finer pores cost more flow than they gain in pumping, that a biporous wick could separate the two, and that the vapour core, through its 1/h³ resistance, decides whether a thin chamber still beats copper.”

Be ready to explain every formula yourself. The model and scripts were built with Claude's help in this project, so before the talk work through Part B with a calculator until you can reproduce the worked examples. The slides' speaker notes also hold each formula's derivation. If your course asks for a declaration on the use of AI tools, that is for you to check and make.